TateConjecture: Lean 4 Structural Specification of the Tate Conjecture for Galois-Invariant Etale Cycle Classes

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/TateConjecture
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Abstract

Standalone Lean 4 structural specification (‘TateConjecture/Basic.lean‘) and comparative documentation (‘docs/STATEMENT.md‘, ‘docs/KNOWN_CASES.md‘, ‘docs/COMPARISON.md‘) of the Tate Conjecture relating Galois-invariant l-adic etale cohomology classes in H^{2r}_et(X_{k^s}, Q_l(r))^{Gal(k^s/k)} to Q_l-linear combinations of algebraic cycles on smooth projective varieties over finitely generated fields.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Formalizes the type-theoretic statement skeleton of the Tate Conjecture in Lean 4. Explicitly states that neither the Tate Conjecture nor the Hodge Conjecture is proved here.

- **Repository:** `https://github.com/BenFrohman/TateConjecture`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652980`
- **Earliest Git Commit SHA:** `9001be1c3db560c2820f1835cdbd81396f618474` (2026-09-21 03:26:04 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

Author: Benjamin Stanley Frohman ([@BenFrohman](https://github.com/BenFrohman)).
Copyright (c) 2026 Benjamin Stanley Frohman. Apache-2.0.
This repository is **not** a proof of the Tate conjecture.
It is **not** a proof of the Hodge conjecture.
It names the sentence. It does not discharge it.
Let ‘X‘ be a smooth projective variety over a finite field ‘F_q‘.
Tate (in codimension ‘i‘) asks that algebraic cycle classes over ‘Q_ ‘
span the Galois invariants in étale cohomology:
CH^i(X)_{Q_ } → H^{2i}_et(X_bar, Q_ (i))^{Gal}
is surjective.
For a fourfold and ‘i = 2‘, that is surfaces spanning the Gal-fixed

part of ‘H^4_et(Q_ (2))‘.
An eigenvalue ‘q^2‘ of geometric Frobenius on ‘H^4_et‘ is the numerical condition for a Tate class of weight 4, twist 2. It is not a surface.
- Does not construct a Galois-fixed class from a complex ‘(2,2)‘-form.
- Does not set the sentence equal to ‘True‘.
- Does not import into [HODGE](https://github.com/BenFrohman/HODGE).
- Does not use a “Frohmanian curve” as a cycle.

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/TateConjecture` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTI` and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	492 B	`e2e4637b6aa6745c...`
`README.md`	1268 B	`906071fc7a279e74...`
`TateConjecture/Basic.lean`	1419 B	`9d5bfe9f9e98c9b8...`
`docs/STATEMENT.md`	747 B	`c5a57b52101ce74e...`

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